

Since the team works with MongoDB Atlas the database is automatically chosen in NoSQL but we will also offer what that database would look like if it were in SQL.

NoSQL Version (The version we use)

```
{
  "_id": ObjectId("65a1b2c3d4e5f6a7b8c9d0e1"), // Auto-generated unique ID (like primary key)
  "email": "player@example.com", // Unique index
  "password": "$2b$10$hashedpassword...", // Bcrypt hash
  "heroName": "SquatMaster",

  // Embedded document - stats
  "stats": {
    "strength": 15,
    "stamina": 12,
    "agility": 8,
    "health": 85
  },

  "level": 5,
  "xp": 450,
  "gold": 250,

  // Embedded array - water intake records
  "waterIntake": [
    {
      "_id": ObjectId("65a1b2c3d4e5f6a7b8c9d0e2"), // Auto-generated sub-ID
      "date": "2024-03-15",
      "cups": 6
    },
    {
      "date": "2024-03-14",
      "cups": 8
    }
  ],

  // Embedded array - workout history
  "workouts": [
    {
      "_id": ObjectId("65a1b2c3d4e5f6a7b8c9d0e3"), "name": "Squats",
      "reps": 30, // Total reps (10 reps × 3 sets)
      "weight": 45,
      "xp": 45,
      "date": ISODate("2024-03-15T10:30:00Z")
    }
  ],

  // Embedded array - battle history
  "battles": [
    {
```

```

    "_id": ObjectId("65a1b2c3d4e5f6a7b8c9d0e4"), "enemyName": "The Lazy Dragon", "enemyHP":
50,
    "enemyMaxHP": 50,
    "enemyDamage": 8,
    "userHP": 85,
    "userMaxHP": 100, "victory": true, "fled": false, "completed": true,
    "date": ISODate("2024-03-15T11:45:00Z")
  }
],

// Embedded array - active quests "activeQuests": [
{
  "_id": ObjectId("65a1b2c3d4e5f6a7b8c9d0e5"), "title": "First Steps",
  "description": "Complete 10 squats", "type": "workout",
  "requirement": 10,
  "completed": 8,          // Current progress "reward": {
  "xp": 50,
  "gold": 25
  }
}
],

// Embedded array - inventory "inventory": [
{
  "item": "Health Potion", "quantity": 3
}
],

```

SQL Version

Table: Users

Column	Type	Constraints
user_id	INT	PRIMARY KEY AUTO_INCREMENT
email	VARCHAR(255)	UNIQUE NOT NULL
password_hash	VARCHAR(255)	NOT NULL
Hero_name	VARCHAR(100)	NOT NULL
level	INT	DEFAULT 1
xp	INT	DEFAULT 0
Gold	INT	DEFAULT 100
Created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

Table: Stats

Column	Type	Constraints
stat_id	INT	PRIMARY KEY AUTO_INCREMENT
user_id	INT	FOREIGN KEY REFERENCES Users(user_id)
strength	INT	DEFAULT 5
stamina	INT	DEFAULT 5
agility	INT	DEFAULT 5
health	INT	DEFAULT 100
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE

Table: Workouts

Column	Type	Constraints
workout_id	INT	PRIMARY KEY AUTO_INCREMENT
user_id	INT	FOREIGN KEY REFERENCES Users(user_id)
name	VARCHAR(100)	NOT NULL
reps	INT	NOT NULL
weight	INT	DEFAULT 0
xp_earned	INT	NOT NULL
workout_date	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

Table: Battles

Column	Type	Constraints
Battle_id	INT	PRIMARY KEY AUTO_INCREMENT
user_id	INT	FOREIGN KEY REFERENCES Users(user_id)
enemy_name	VARCHAR(100)	NOT NULL
victory	BOOLEAN	DEFAULT FALSE
damage_dealt	INT	NOT NULL
damage_taken	INT	NOT NULL
xp_reward	INT	DEFAULT 0
gold_reward	INT	DEFAULT 0
Battle_date	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

Table: Water_Intake

Column	Type	Constraints
water_id	INT	PRIMARY KEY AUTO_INCREMENT
user_id	INT	FOREIGN KEY REFERENCES Users(user_id)
Date	DATE	NOT NULL
cups	DECIMAL(3,1)	NOT NULL

Table: Quests

Column	Type	Constraints
quest_id	INT	PRIMARY KEY AUTO_INCREMENT
Title	VARCHAR(200)	NOT NULL
description	TEXT	

type	ENUM('workout','hydration','boss')	NOT NULL
Requirement	INT	NOT NULL
xp_reward	INT	NOT NULL
Gold_reward	INT	NOT NULL

Table: User_Requests (Many-to-Many relationship)

Column	Type	Constraints
user_request_id	INT	PRIMARY KEY AUTO_INCREMENT
user_id	INT	FOREIGN KEY REFERENCES Users(user_id)
request_id	INT	FOREIGN KEY REFERENCES Requests(request_id)
progress	INT	DEFAULT 0
Completed	BOOLEAN	DEFAULT FALSE
claimed	BOOLEAN	DEFAULT FALSE
assigned_date	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

Table: Inventory

Column	Type	Constraints
inventory_id	INT	PRIMARY KEY AUTO_INCREMENT
user_id	INT	FOREIGN KEY REFERENCES Users(user_id)
item_name	VARCHAR(100)	NOT NULL
quantity	INT	DEFAULT 1